CIS 2267 Linear Regression

Data Source Attribution and Citation

Source content for this assignment is "Student projects for *Murach's Python for Data Analysis*" made available by the publisher for courses using this textbook. Source data is provided by the Seaborn library maintainers.

Guidelines

- Start your Notebook by importing any modules, classes, or functions required by the project.
- Use one code cell to store the code that accomplishes each task and question.
- Use comments to describe your thought process and to document any code that's hard to understand.
- Use Markdown cells to create headings that organize your Notebooks.

Dataset Summary

This dataset contains information about diamond prices based on carat weight, cut quality, color, clarity, depth percent, and table percent. This dataset also includes x, y, and z columns that represent the diamond's length, width, and depth.

Your Tasks

The Seaborn library maintainers provide sample datasets to help learners explore the data analysis and visualization capabilities of Seaborn. To make things simpler for users, there is a **`seaborn.load_dataset`** function that automatically downloads named datasets without worrying about URLs, hyperlinks, or saving files from the internet.

1. Read the data file from the Seaborn website. To load this data into a **`DataFrame`**, run this code:

   ```
   data = sns.load_dataset('diamonds')
   ```

2. Display the first five rows of data.
3. Use a condensed heatmap to identify correlations for the price column. Sort the results, include annotations that format the values with three decimal places, and remove the color bar.
4. Use the column with the strongest correlation to split the data into test and training datasets, where the test dataset consists of 30% of the total dataset. Be sure to specify a value for the **`random_state`** parameter.
5. Create a **`LinearRegression`** object, and then fit the training dataset to the model.
6. Score the model using the test dataset.
7. Predict the y values based on the x values in the test dataset, and store the results in a variable.
8. Put the predicted values in a **`DataFrame`**.
9. Join the **`y_test`** and predicted data with the **`x_test`** data, save the joined data in a new **`DataFrame`**, and then display the first five rows of data.
10. Melt the actual and predicted price columns together, assigning appropriate values to the variable and value names.
11. Use a Seaborn line plot without a confidence interval to plot the predicted data.